

ASYMPTOTICS OF PARTITION FUNCTIONS OF COULOMB GASES

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Dissertation's defence for the degree of Master of Sciences in Mathematics.
Supervisor: Guilherme Silva



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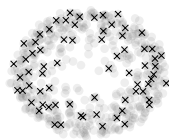
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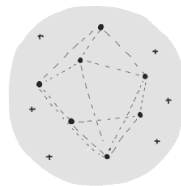
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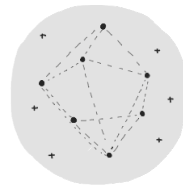
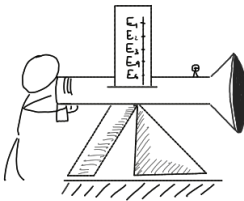
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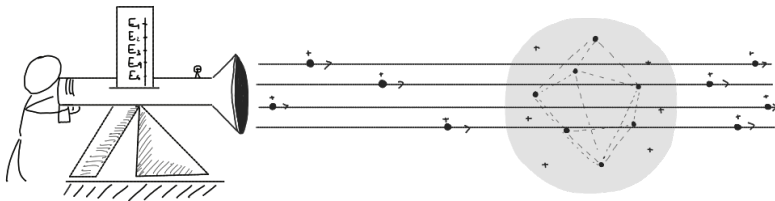
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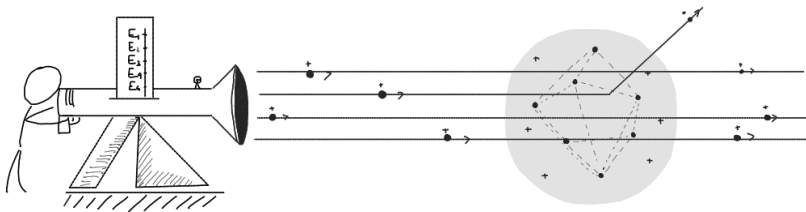


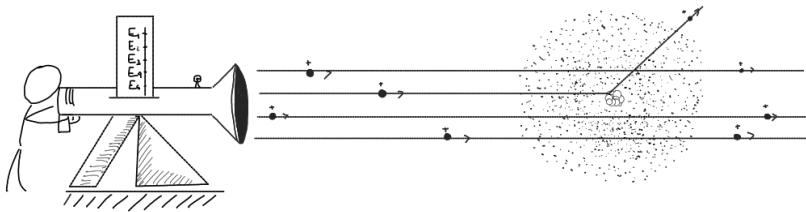
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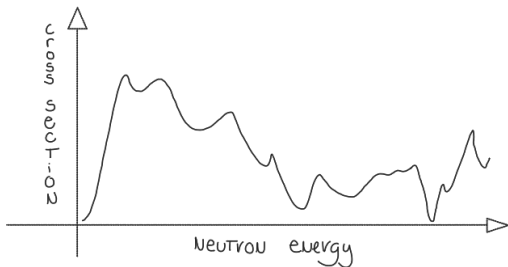
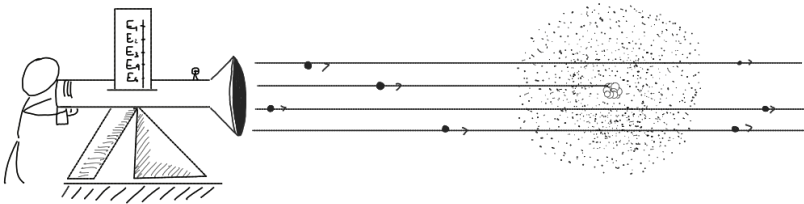


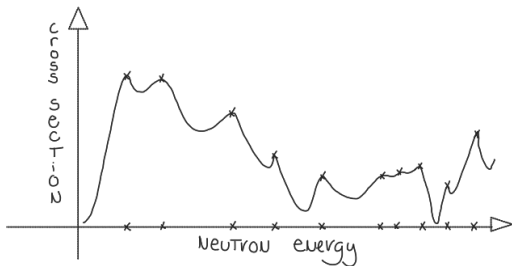
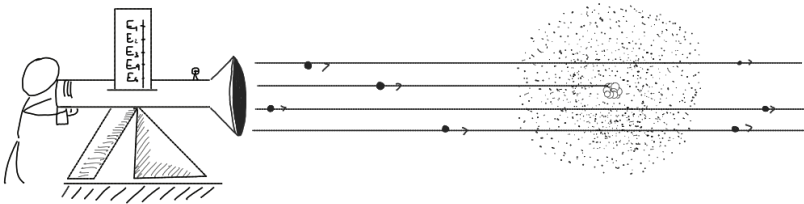






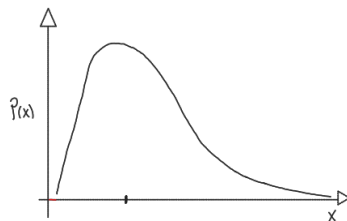
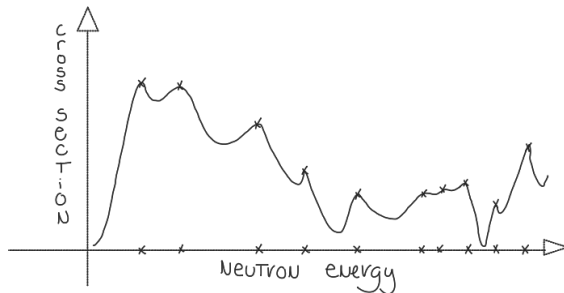






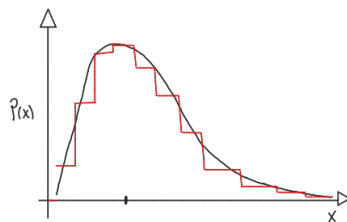
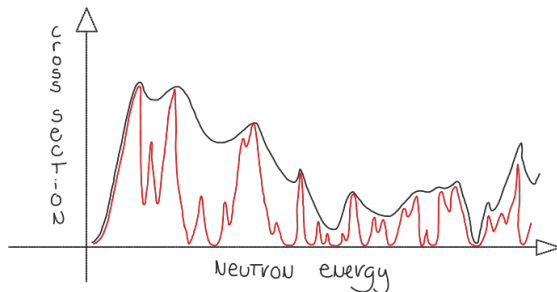
EUGENE WIGNER - *On the Wigner Surmise* (1956)

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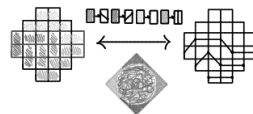
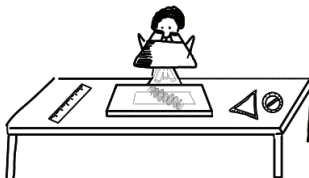
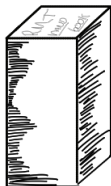
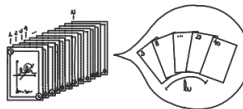
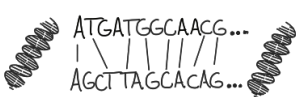
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A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} & x_{1,4} \\ x_{2,1} & x_{2,2} & x_{2,3} & x_{2,4} \\ x_{3,1} & x_{3,2} & x_{3,3} & x_{3,4} \\ x_{4,1} & x_{4,2} & x_{4,3} & x_{4,4} \\ x_{5,1} & x_{5,2} & x_{5,3} & x_{5,4} \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
Discrete	$\mathbb{R}$	General
Uniform	$\mathbb{C}$	Hermitian
Gaussian	$\mathbb{H}$	Normal
$\vdots$	$\vdots$	$\vdots$

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A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -1 & -1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
Discrete	$\mathbb{R}$	General
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$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^n p(x_{i,j}) = U^{n^2}(\{-1, 1\})$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} 0.213 & 0.368 & 0.912 & 0.881 \\ 0.759 & 0.692 & 0.000 & 0.332 \\ 0.898 & 0.987 & 0.422 & 0.623 \\ 0.145 & 0.880 & 0.522 & 0.686 \\ 0.753 & 0.687 & 0.787 & 0.515 \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
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Uniform	$\mathbb{C}$	Hermitian
Gaussian	$\mathbb{H}$	Normal
$\vdots$	$\vdots$	$\vdots$



$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^n p(x_{i,j}) = U^{n^2}([0, 1])$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -0.022 & -1.340 & -0.483 & -1.134 \\ 1.752 & -1.644 & 1.081 & -0.936 \\ -1.058 & -0.091 & 0.544 & -0.382 \\ 0.633 & 0.567 & 0.606 & -0.621 \\ -0.316 & -1.128 & -0.150 & 0.251 \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
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Uniform	$\mathbb{C}$	Hermitian
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$\vdots$	$\vdots$	$\vdots$



Ginibre Orthogonal Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^n e^{-n x_{i,j}^2}$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -2.0+0.1i & -0.2-0.9i & -0.4-0.3i & -0.3+1.0i \\ -1.6-0.8i & 0.3-0.6i & -1.7+0.9i & -0.1+0.3i \\ -0.5-0.9i & -0.5+0.2i & 1.2+1.0i & 0.4-1.3i \\ -0.6+2.4i & 0.1-0.7i & -0.2+1.6i & 0.6-0.4i \\ 0.1-1.3i & -0.2-0.9i & 1.2-1.6i & 2.1+2.0i \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
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$\vdots$	$\vdots$	$\vdots$



Ginibre Unitary Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^n e^{-nx_{i,j}^2} e^{-ny_{i,j}^2}$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -0.8 & -0.2-1.5i & 0.9-0.6i & -3.2+2.3i \\ -0.2+1.5i & 0.5 & -1.8-0.8i & 1.5+1.8i \\ 0.9+0.6i & -1.8+0.8i & 1.8 & 2.1+2.5i \\ -3.2-2.3i & 1.5-1.8i & 2.1-2.5i & 2.4 \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
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Gaussian	$\mathbb{H}$	Normal
$\vdots$	$\vdots$	$\vdots$



Gaussian Unitary Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^n e^{-nz_{i,j}^2} \prod_{i>j}^n e^{-2n|z_{i,j}|^2}$$

We can think of the matrix itself as a random variable, a measurable map from a probability measure space to $\mathcal{M}_n$.

$$\begin{bmatrix} -0.8 & -0.2-1.5i & 0.9-0.6i & -3.2+2.3i \\ -0.2+1.5i & 0.5 & -1.8-0.8i & 1.5+1.8i \\ 0.9+0.6i & -1.8+0.8i & 1.8 & 2.1+2.5i \\ -3.2-2.3i & 1.5-1.8i & 2.1-2.5i & 2.4 \end{bmatrix}$$

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Gaussian Unitary Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^n e^{-nz_{i,j}^2} \prod_{i>j}^n e^{-2n|z_{i,j}|^2} \propto e^{-n\text{tr}(\hat{X}^2)}$$

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We can think of the matrix itself as a random variable, a measurable map from a probability measure space to $\mathcal{M}_n$.

Normal matrices are similar to diagonal matrices, i.e.

$$\hat{X} = \hat{U} \hat{\Lambda} \hat{U}^*.$$

Moreover, the spectra is typically simple.

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$$\hat{X} \mapsto (\hat{\Lambda}, \hat{U}) = \left(\begin{array}{c} \begin{array}{c} \text{Scatter plot of eigenvalues in the complex plane with axes } \mathbb{R} \text{ and } i\mathbb{R} \end{array}, \begin{array}{c} \text{Diagram of eigenvectors represented as arrows originating from a single point} \end{array} \end{array} \right),$$

We can think of the matrix itself as a random variable, a measurable map from a probability measure space to $\mathcal{M}_n$.

Invariance by unitary conjugation means independence of coordinates.

$$d\mathcal{P}(\hat{X}_1) = d\mathcal{P}(\hat{X}_2) \iff \hat{X}_1 = \hat{U}\hat{X}_2\hat{U}^*$$

for some unitary U .

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$$\hat{X} \mapsto (\hat{\Lambda}, \hat{U}) = \left(\begin{array}{c} \text{Scatter plot of eigenvalues } \lambda_i \text{ in the } \mathbb{R} \text{ vs } i\mathbb{R} \text{ plane} \\ \text{Diagram of eigenvectors } u_i \text{ as arrows originating from a point} \end{array} \right),$$

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We want that $d\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{X})d\hat{X}$ is invariant.

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$$\begin{aligned} \exp\left(-n\text{tr}\left\{Q(\hat{U}\hat{X}_1\hat{U}^*)\right\}\right) &= \exp\left(-n\text{tr}\left\{Q(\hat{U}^*\hat{U}\hat{X}_1)\right\}\right) = \exp\left(-n\text{tr}\left\{Q(\hat{X}_1)\right\}\right) \\ &= \exp\left(-n\text{tr}\left\{Q(\hat{U}\hat{\Lambda}\hat{U}^*)\right\}\right) = \exp\left(-n\sum_i^n Q(\lambda_i)\right). \end{aligned}$$

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We want that $d\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{X})d\hat{X}$ is invariant.

$$d\hat{X} = \prod_{1 \leq i < j \leq n} |z_i - z_j|^2 \prod_{i=1}^n d^2 z_i d\hat{U},$$

where $d\hat{U}$ is the Haar measure on eigenvectors and $d^2 z$ is Lebesgue measure on $\mathbb{C}$.

We set the ensemble

$$\mathcal{E} = (\mathcal{S}, \mathcal{P}) \quad \text{with} \quad \begin{cases} \mathcal{S} = \mathcal{N}_n(\mathbb{C}) \\ \mathcal{P}(\hat{X}) = \exp \left\{ -n \operatorname{tr} \left(Q(\hat{X}) \right) \right\} \end{cases}$$

such that

$$d\mathcal{P}(\hat{X}) = \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)} d^2 z_j d\hat{U}.$$

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$$\begin{aligned} \int_{\mathcal{M}_n} d\mathcal{P}(\hat{X}) &= \int_{\mathbb{C}^n} \int_{\tilde{U}(n)} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)} d^2 z_j d\hat{U} \\ &= \int_{\mathbb{C}^n} \frac{\prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)}}{\mathcal{Z}_n^Q} d^2 z_j \end{aligned}$$

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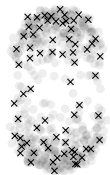
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we have

$$p_n(\{z_i\}_{i=1}^n) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)}$$



MODELING OUR MODEL

We know that

$$p_n(\{z_i\}_{i=1}^n) = p\left(\begin{array}{c} \text{Diagram of } n \text{ points in } \mathbb{R}^2 \\ \text{with axes labeled } \mathbb{R} \end{array}\right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)}$$

We know that

$$p_n(\{z_i\}_{i=1}^n) = p\left(\begin{array}{c} \mathbb{R}^2 \\ \text{cloud of points} \\ \mathbb{R} \end{array}\right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)}$$

We can notice two effects, one coming from the Jacobian and the other one from Q .

$$p_n\left(\begin{array}{c} \mathbb{R}^2 \\ \text{cloud of points} \\ \mathbb{R} \end{array}\right) > \left\{ \right.$$

We know that

$$p_n(\{z_i\}_{i=1}^n) = p \left(\text{Diagram of } n \text{ points in the complex plane} \right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)}$$

We can notice two effects, one coming from the Jacobian and the other one from Q .

$$p_n \left(\text{Diagram of } n \text{ points in the complex plane} \right) > \left\{ p_n \left(\text{Diagram of } n \text{ points in the complex plane} \right), \text{ the interactions dominates.} \right.$$

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We can notice two effects, one coming from the Jacobian and the other one from Q .

$$p_n \left(\text{Diagram of } n \text{ points in the complex plane} \right) > \begin{cases} p_n \left(\text{Diagram of } n \text{ points clustered in the center} \right), & \text{the interactions dominates.} \\ p_n \left(\text{Diagram of } n \text{ points spread out} \right), & \text{the external potential dominates.*} \end{cases}$$

We know that

$$p_n(\{z_i\}_{i=1}^n) = p\left(\begin{array}{c} \mathbb{R} \\ \text{Diagram of } n \text{ points in the plane} \\ \mathbb{R} \end{array}\right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)}$$

We know that

$$\begin{aligned} p_n(\{z_i\}_{i=1}^n) &= p\left(\begin{array}{c} \mathbb{R} \\ \text{Diagram of } n \text{ points in } \mathbb{R}^2 \\ \mathbb{R} \end{array}\right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)} \\ &= \frac{1}{\mathcal{Z}_n^Q} \exp \left\{ -n^2 \left(\frac{1}{n} \sum_{i=1}^n Q(z_i) + \frac{1}{n^2} \sum_{i \neq j} \ln |z_i - z_j| \right) \right\} \end{aligned}$$

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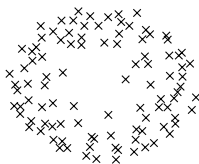
$$\begin{aligned}
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The measure dp_n models an Coulomb-interacting gas of charged indistinguishable particles under some external field Q on $\mathbb{C}$.

If we want to study this gases, how to describe its statistical properties?

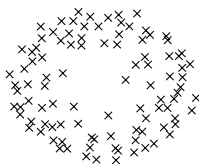
POINT PROCESS

Suppose there are n particles at random locations in the complex plane $\mathbb{C}$ denoted by the configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$.



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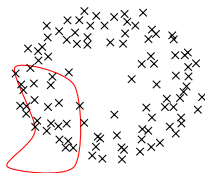


Define a quantity $\xi(A)$ s.t.

$$\xi(A) = \sum_{k=1}^n \delta(z_k)$$

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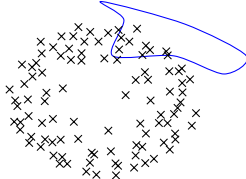


Define a quantity $\xi(A_1)$ s.t.

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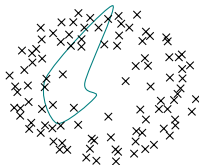


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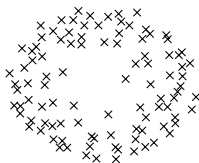


Define a quantity $\xi(A_3)$ s.t.

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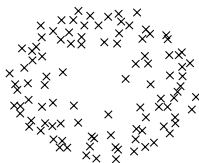
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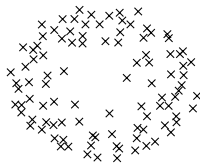


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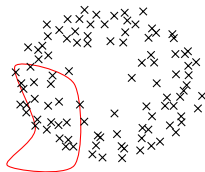
$$\xi(A) = \sum_{k=1}^n \delta(z_k)$$

This defines a *random counting measure* in the usual sense. A **point process** $\mathbb{P}$ on $\mathbb{C}$ is a probability measure on the space of all counting measures $\mathcal{N}(\mathbb{C})$.

Consider the configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$.

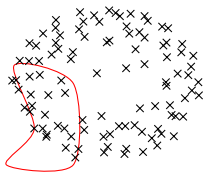


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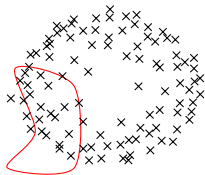
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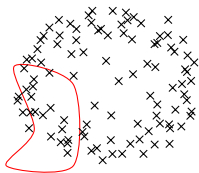
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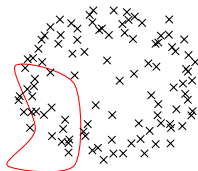
$$\xi^{(2)}(A_1) = \sum_{k=1}^n \delta_{A_1}(z_k^{(2)})$$

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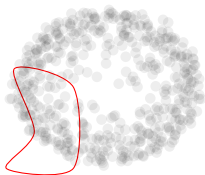


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We define

$$p(A) := \mathbb{E}[\xi(A)] = \mathbb{E}[\#(\chi_n \cap A)], \quad A \in \mathcal{E}.$$

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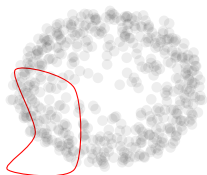
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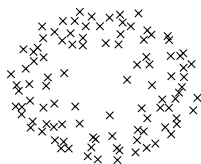
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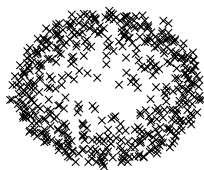
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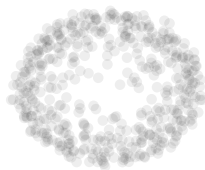
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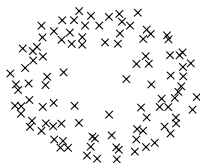
where d^2z is the usual area measure. The **1-point correlation function** p_1 represents the infinitesimal probability of a point of the process being at a location z , i.e. $\mathbb{P}(\xi(d^2z) = 1) = p_1(z)d^2z$.







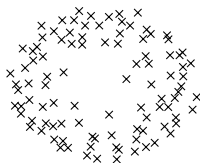
VARIATIONAL PROBLEM



$$p \left(\begin{array}{c} \text{circle of points} \\ \text{with axes labeled } \mathbb{R} \text{ and } \mathbb{R} \end{array} \right) = \frac{e^{-n^2 \mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q},$$

$$\mathcal{H}(\chi_n) = \frac{1}{n^2} \sum_{i \neq j}^n \ln |z_i - z_j| + \frac{1}{n} \sum_{i=1}^n Q(z_i).$$

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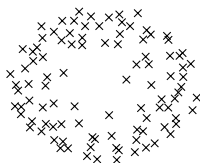


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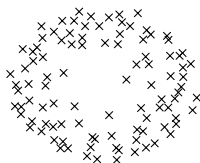


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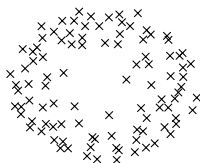


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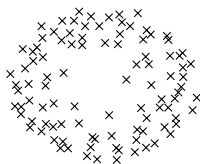


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Associated with the variational problem defined by

$$E_Q = \inf_{\mu \in \mathcal{M}^1} \left[\int \int_{t \neq s} \ln |t - s|^{-1} d\mu(t) d\mu(s) + \int Q(s) d\mu(s) \right] = I_Q(\mu_Q),$$

one defines the so-called *Euler-Lagrange* type variational equations.

THEOREM

(Strong Variational Equations) Suppose that $d\mu_Q = \mu(z)d^2z$ for some continuous function μ of compact support. The equilibrium measure μ_Q for the given variational problem, where $\Sigma = \mathbb{C}$ satisfies the following conditions. There exists a real constant l such that

$$\textcircled{1} \quad 2 \int \ln |z - t|^{-1} d\mu_Q(t) + Q(z) \geq l$$

for all $z \in \mathbb{C}$.

$$\textcircled{2} \quad 2 \int \ln |z - t|^{-1} d\mu_Q(t) + Q(z) = l$$

on $\text{supp}(\mu)$.

We define $U^{\mu_Q}(z) = \int \ln |z - t|^{-1} d\mu_Q(t)$.

NORMAL ENSEMBLE - ROTATIONAL SYMMETRY

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- 3 Furthermore, assume Q to be smooth, sub-harmonic in $\mathbb{C}$, and strictly sub-harmonic in a neighborhood of the $\text{supp}(\mu_Q)$.

COROLLARY

If r_0 is the largest solution of $rq'(r) = 0$ and r_1 is the smallest solution of $rq'(r) = 2$ then the support of μ_Q is given by the ring $S = \{z | r_0 \leq |z| \leq r_1\}$ and $d\mu_Q$ is given by

$$d\mu_Q(z) = \frac{1}{4\pi} (rq'(r))' \delta_S dr d\phi, \quad z = r e^{i\phi}.$$

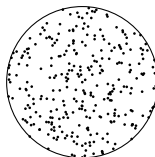
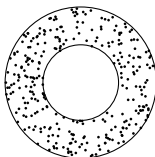
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The set $\mathcal{S}$ can be either an annulus (if $r_0 > 0$) or a disk (if $r_0 = 0$).



We call **Droplet** the support of μ_Q , which in our case is the set of all complex values z such that $r_0 \leq |z| \leq r_1$.

Let $\mu \in \mathcal{M}^1(\mathbb{C})$ be a measure such that $d\mu = \frac{1}{4\pi}(rq'(r))'\delta_S d\phi dr$.

$$\int_{\mathbb{C}} d\mu = 1,$$

that is,

$$r_1 q'(r_1) - r_0 q'(r_0) = 2.$$

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$$2U^\mu(z) \geq I - Q(z), \quad z \in \mathbb{C}$$

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with $I_Q(\mu) < \infty$, then $\mu_Q = \mu$. We must compute

$$U^\mu(z) = \int \ln |z - t|^{-1} d\mu_Q(t) = \frac{1}{4\pi} \int_{r_0}^{r_1} (rq'(r))' \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\phi}|} d\phi dr.$$

With the formula

$$\int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\phi}|} d\phi = \begin{cases} 2\pi \ln \frac{1}{r}, & |z| \leq r, \\ 2\pi \ln \frac{1}{|z|}, & |z| > r. \end{cases}$$

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for $r_0 \leq |z| \leq r_1$

$$\begin{aligned} U^\mu(z) &= \frac{1}{2} \int_{r_0}^{|z|} (rq'(r))' \ln \left(\frac{1}{r} \right) dr + \frac{1}{2} \int_{|z|}^{r_1} (rq'(r))' \ln \left(\frac{1}{|z|} \right) dr, \\ &= -\frac{1}{2} (2 \ln(r_1) - q(r_1)) - \frac{1}{2} q(|z|). \end{aligned}$$

Define $I = -(2 \ln(r_1) - q(r_1))$, then

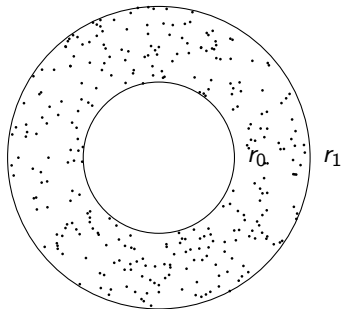
$$2U^\mu(z) = \begin{cases} I - q(r_0), & |z| < r_0, \\ 2 \ln \frac{1}{|z|}, & |z| > r_1, \\ I - q(|z|), & r_0 \leq |z| \leq r_1. \end{cases}$$

We need to prove that for $z \in \mathbb{C}$, we have $2U^\mu(z) \geq I - q(|z|)$ while on $r_0 \leq |z| \leq r_1$ we have $2U^\mu(z) = I - q(|z|)$.

Define $l = -(2\ln(r_1) - q(r_1))$, then

$$2U^\mu(z) = \begin{cases} l - q(r_0), & |z| < r_0, \\ 2\ln \frac{1}{|z|}, & |z| > r_1, \\ l - q(|z|), & r_0 \leq |z| \leq r_1. \end{cases}$$

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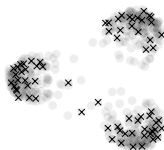


r_0 is the biggest r s.t. $rq'(r) = 0$.

$$q(r_0) \leq q(|z|), \quad |z| < r_0.$$

r_1 is the smallest r s.t. $rq'(r) = 2$. Then, for $|z| > r_1$

$$\begin{aligned} q(|z|) - 2\log(|z|) &\geq q(r_1) - 2\log(r_1) \\ \implies 2\ln \frac{1}{|z|} &\geq l - q(|z|) \end{aligned}$$



ASYMPTOTICS OF THE PARTITION FUNCTION

QUICK RECAP

$$d\mathcal{P}(\hat{X}) = \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_j)} d^2 z_i d\hat{U}.$$

$\mathcal{P}(\hat{X})$	$\mathbb{S}$	$\mathcal{M}_n$
$e^{-n \operatorname{tr}(\hat{M}^2)}$	$\mathbb{R}$	General
$e^{-n \operatorname{tr}\{(\hat{X}^* \hat{X})\}}$	$\mathbb{C}$	Hermitian
$e^{-n \operatorname{tr}\{Q(\hat{X})\}}$	$\mathbb{H}$	Normal
$\vdots$	$\vdots$	$\vdots$



Orthogonal Polynomial Ensemble

QUICK RECAP

$$p_n(\{z_i\}_{i=1}^n) = \frac{1}{Z_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_i)}$$

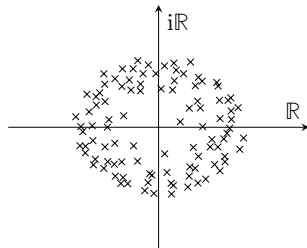
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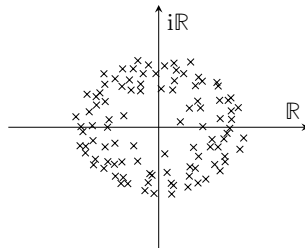
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QUICK RECAP

$$p_n(\{z_i\}_{i=1}^n) = \frac{e^{-n^2 \mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q}, \quad \text{with}$$

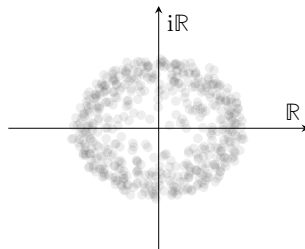
$$\mathcal{H}(\chi_n) = \frac{1}{n} \sum_{i=1}^n Q(z_i) + \frac{1}{n^2} \sum_{i \neq j}^n \ln |z_i - z_j|.$$



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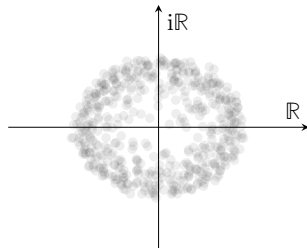


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$$U^\mu(z) := \int \ln \frac{1}{|z - t|} d\mu(t),$$

$$I_Q(\mu) := \int U^\mu(s) d\mu(s) + \int Q(s) d\mu(s),$$

μ_Q minimizer of I_Q .

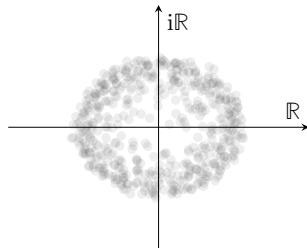


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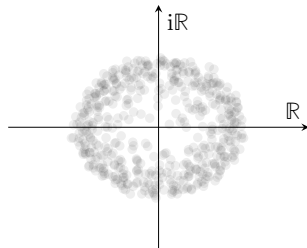
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$$\ln(\mathcal{Z}_n^Q) = -I_Q n^2 + \frac{1}{2} n \ln n + E_Q n + G \ln n + F_Q + o(1), \quad n \rightarrow \infty.$$

Result derived in Byun, Sung-Soo and Kang, Nam-Gyu and Seo, Seong-Mi article.

SKETCH

- ① *Partition functions can be expressed in terms of the **norming constants**: We can write the free energy as*

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

- ② *Asymptotics of norming constant using the asymptotic **Laplace technique**.*

$$h_j = \int_{\mathbb{C}} |z|^{2j} e^{-nQ(z)} d^2z \approx \frac{e^{-nV_{\tau(j)}(r_c(\tau(j)))}}{\sqrt{n}}.$$

- ③ *Asymptotic of the summation on $\ln h_j$ using **quadrature techniques**:*

$$-\sum_{j=0}^{n-1} \ln h_j \approx -n^2 \int_{r_0}^{r_1} V(r) \frac{(rq'(r))'}{2} dr = -n^2 I_Q[\mu_Q].$$

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$$\frac{\prod_{i=1}^n w(z_i)}{\prod_{l=1}^n h_l} \prod_{i < j}^n |z_i - z_j|^2 = \det \left(\sqrt{w(z_i)w(z_j)} \sum_{l=1}^n \frac{1}{h_l} \overline{q_{l-1}(z_i)} q_{l-1}(z_j) \right)_{n \times n}$$

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THEOREM

Let $\{q_j\}_{j=0}^\infty$ be a family of monic polynomials such that each $q_j(\lambda)$ is of degree j . Impose also that exists finite positive constants $\{h_{j,n}\}_{j=0}^\infty$ s.t.

$$\langle q_j, q_k \rangle_w = \int_{\mathbb{C}} q_j(z) \overline{q_k(z)} e^{-nQ(z)} d^2z = h_j \delta_{\{j=k\}}$$

for any $j, k > 0$. Then, $K_n(x, y)$ has the reproducing property, i.e.

$$\int_{\mathbb{C}} \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^{k+1} d^2\lambda_{k+1} = (n - k) \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^k.$$

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COROLLARY

Because of

$$\int_{\mathbb{C}} \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^{k+1} d^2 \lambda_{k+1} = (n - k) \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^k,$$

the partition function $\mathcal{Z}_n^Q$ can be written as

$$\mathcal{Z}_n^Q = \prod_{l=1}^n h_l \int_{\mathbb{C}^n} \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^n \prod_{i=1}^n d^2 \lambda_i = n! \prod_{l=1}^n h_l$$

NORMING CONSTANTS

We want to find

$$\int_{\mathbb{C}} P_k(x) \overline{P}_m(x) w(x) d^2 z = \langle P_k, P_m \rangle_w = \begin{cases} h_m & \text{if } m = k, \\ 0 & \text{if } m \neq k. \end{cases}$$

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$$\begin{aligned} \langle z^m, z^j \rangle_w &= \int_{\mathbb{C}} z^m \bar{z}^j e^{-nQ(z)} d^2 z, \\ &= \int_0^\infty \int_0^{2\pi} r^{m+j} e^{i\theta(m-j)} e^{-nq(r)} \frac{r}{\pi} dr d\theta, \\ &= \frac{1}{\pi} \int_0^{2\pi} e^{i\theta(m-j)} d\theta \int_0^\infty r^{m+j+1} e^{-nq(r)} dr \end{aligned}$$

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NORMING CONSTANTS

Defining $z = r e^{i\theta}$ ($|z| = r$), because of the symmetry on Q we write

$$\begin{aligned}
 h_j &= \int_{\mathbb{C}} |z|^{2j} e^{-nQ(z)} d^2z \\
 &= \int_{\Theta} \int_{\mathbb{R}_+} r^{2j} e^{-nq(r)} \frac{r}{\pi} dr d\theta, \\
 &= 2 \int_{\mathbb{R}_+} r^{2j+1} e^{-nq(r)} dr, \\
 &= \int_{\mathbb{R}_+} 2r e^{-n(q(r) - \frac{2j}{n} \ln(r))} dr.
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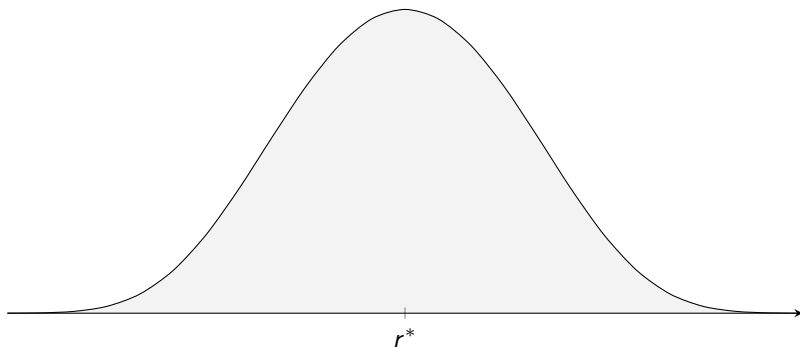
We redefine our potential using

$$V_{\tau(j)}(r) = q(r) - 2\tau(j) \ln(r),$$

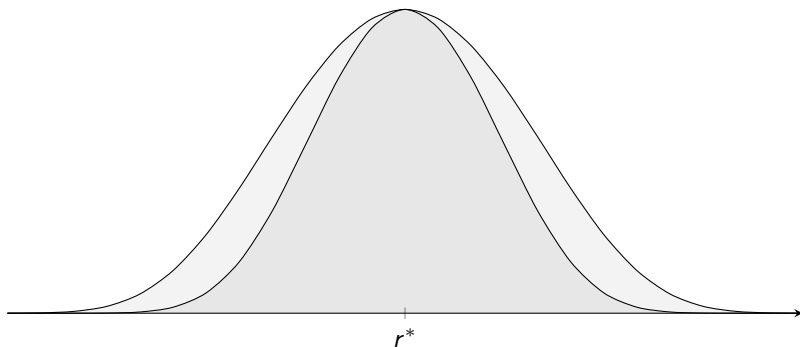
where $\tau(j) := j/n \in [0, 1)$. Then, rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r e^{-nV_{\tau(j)}(r)} dr.$$

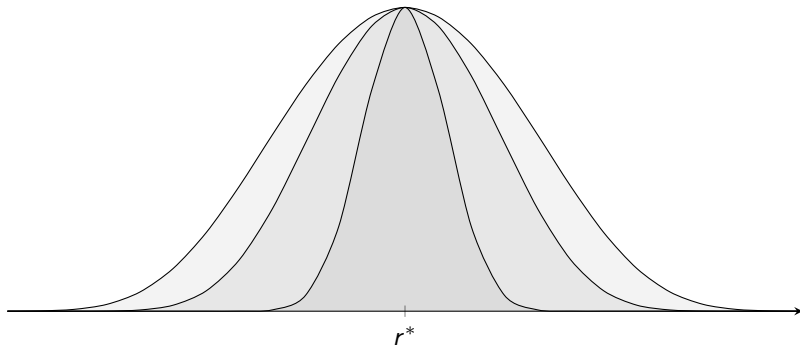
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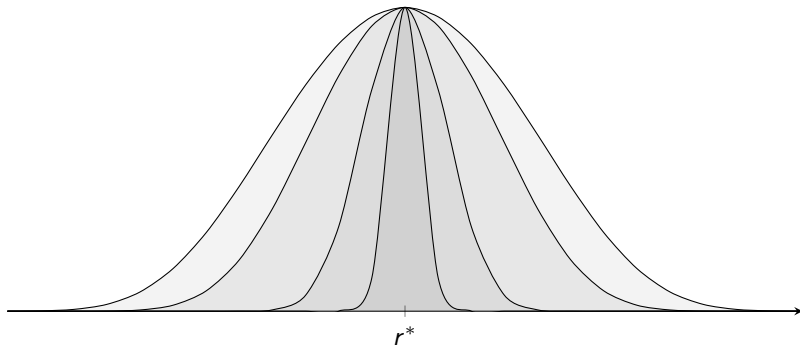
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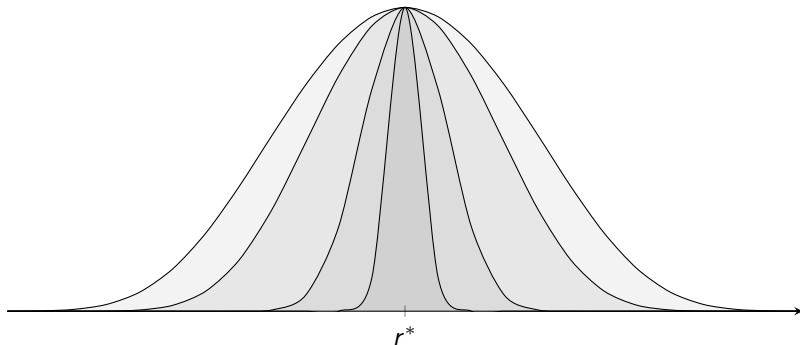
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$$\lim_{n \rightarrow \infty} \frac{e^{-n(V_{\tau(j)}(r^*))}}{e^{-n(V_{\tau(j)}(r^* - \epsilon))}} = \lim_{n \rightarrow \infty} e^{-n(V_{\tau(j)}(r^*) - V_{\tau(j)}(r^* - \epsilon))} = \lim_{n \rightarrow \infty} e^{n|c|} \rightarrow +\infty.$$

THEOREM

(Laplace's Method) Let Φ be

$$\Phi(n) = \int_a^b g(t) e^{-n\phi(t)} dt$$

where $g(t) e^{-n\phi(t)}$ is absolutely integrable over $[a, b]$ for every n , the function $\phi(t)$ attains its maximum only at the point c in (a, b) , the second derivative $\phi''(t)$ exists in a neighborhood of and is negative on c . Finally, $g(t)$ is non-zero and continuous at c . Then the function Φ satisfies an estimate of the form

$$\Phi(n) = e^{-n\phi(c)} \left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}n^{1/2}} + \mathcal{O}(n^{-3/2}) \right)$$

as $n \rightarrow \infty$.

Taking a look at the first derivative of the newly defined potential we find that the critical point r_c must satisfy

$$V'_\tau(r) = q'(r) - \frac{2\tau}{r} \quad \implies \quad r_c q'(r_c) = 2\tau.$$

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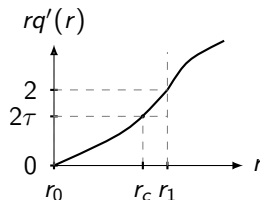
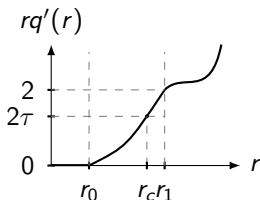
Denote $r_{\tau(j)}$ the r such that $rq'(r) = 2\tau(j)$.

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Denote $r_{\tau(j)}$ the r such that $rq'(r) = 2\tau(j)$. Also define the limit points when $\tau = 0$ and $\tau = 1$ as

$$\begin{cases} r_0, & \text{the largest solution of } rq'(r) = 0, \\ r_1, & \text{the smallest solution of } rq'(r) = 2. \end{cases}$$



THEOREM

We have the following asymptotic of n for h_j , with $0 \leq j \leq n-1$,

$$h_j = \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right],$$

where we have set

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3} \frac{5}{24}.$$

QUADRATURE

We have

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

where

$$h_j = \int_{\mathbb{C}} |z|^{2j} e^{-nQ(z)} d^2z \approx \frac{e^{-nV_{\tau(j)}(r_c(\tau(j)))}}{\sqrt{n}}.$$

QUADRATURE

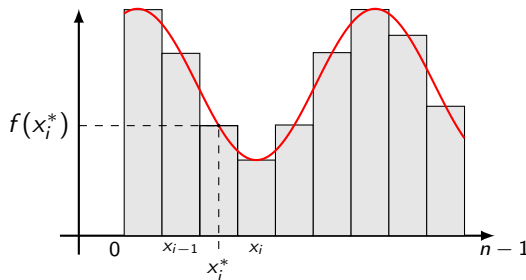
We have

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$$\Delta_n \sum_{j=0}^{n-1} \ln h_j \approx \int_a^b \ln h_j$$



QUADRATURE I

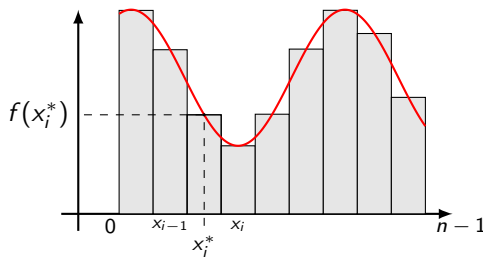
We begin with $\Delta_n = (b - a)/n$, s.t.

$$\begin{aligned}
 \int_a^b f(x) dx &\approx \Delta_n \sum_{j=0}^n \frac{f(a + j\Delta_n) + f(a + (j+1)\Delta_n)}{2}, \\
 &= \Delta_n \left[\frac{f(a)}{2} + f(a + \Delta_n) + \cdots + f(b - \Delta_n) + \frac{f(b)}{2} \right], \\
 &= \Delta_n \sum_{j=1}^{n-1} f(x_j^*) + \frac{\Delta_n}{2} (f(b) + f(a)), \\
 &= \Delta_n \sum_{j=0}^{n-1} f(x_j^*) + \frac{\Delta_n}{2} (f(b) - f(a)).
 \end{aligned}$$

THEOREM

(Euler-Maclaurin Formula) For some function $f \in C^p(\mathbb{R})$ and $p \geq 1$ let $\Delta_n = (b - a)/n$. Then, the Euler-Maclaurin formula for the interval $[a, b]$ is given by

$$\Delta_n \sum_{j=1}^{n-1} f(a + j\Delta_n) = \int_a^b f(x) dx - \frac{\Delta_n}{2} (f(b) - f(a)) + \sum_{r=1}^{p-1} \frac{\Delta_n^{r+1} B_{r+1}}{(r+1)!} [f^{(r)}(b) - f^{(r)}(a)] + R_p$$



We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\sum_{j=0}^{n-1} \ln \left[\frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right] \right],$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\sum_{j=0}^{n-1} \left[-n V_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \left(\ln(2\pi r_{\tau(j)}^2) - \ln n - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right]$$

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$$\sum_{j=0}^{n-1} \left[-nV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \left(\ln(2\pi r_{\tau(j)}^2) - \ln n - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right]$$

THEOREM

$$-n \sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) = -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) + \mathcal{O}(n^{-2}),$$

where

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

$$-2U_{\mu_Q}(0) := \int_{r_0}^{r_1} \ln(s) (sq'(s))' ds = q(r_1) - q(r_0) + 2 \ln(r_1).$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\begin{aligned}
 & -n^2 I_Q[\mu_Q] + n U_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) \\
 & + \sum_{j=0}^{n-1} \left[\frac{1}{2} \left(\ln(2\pi r_{\tau(j)}^2) - \ln n - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right]
 \end{aligned}$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$-n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) + \sum_{j=0}^{n-1} \left[\frac{1}{2} \left(\ln(2\pi r_{\tau(j)}^2) - \ln n - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right]$$

THEOREM

Asymptotically on n ,

$$\begin{aligned} \sum_{j=0}^{n-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) &= nE_Q[\mu_Q] - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + \mathcal{O}(n^{-1}), \\ \sum_{j=0}^{n-1} \ln(2\pi r_{\tau(j)}^2) &= n \ln(2\pi) + 2 \left(-nU_{\mu_Q}(0) - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) + \mathcal{O}(n^{-1}) \right), \\ \frac{1}{n} \sum_{j=0}^{n-1} \mathfrak{B}(r_{\tau(j)}) &= \frac{1}{4} \int_0^{r_1} \mathfrak{B}(s) s V^{(2)}(s) ds + \mathcal{O}(n^{-2}) \end{aligned}$$

where

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (sq'(s))' ds.$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\begin{aligned} \sum_{j=0}^{n-1} \ln h_j &= -n^2 l_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) - \frac{n}{2} \ln(n) - \frac{1}{2} \left(nE_Q[\mu_Q] - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \right) \\ &\quad + \frac{1}{2} \left(n \ln(2\pi) + 2 \left(-nU_{\mu_Q}(0) - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) \right) \right) + \frac{1}{4} \int_0^{r_1} \mathfrak{B}(s) s V^{(2)}(s) ds + \mathcal{O}(n^{-1}) \end{aligned}$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\begin{aligned} \sum_{j=0}^{n-1} \ln h_j &= -n^2 I_Q[\mu_Q] - \frac{n}{2} \ln(n) + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{2}{3} \ln \left(\frac{r_0}{r_1} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \\ &\quad - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + \mathcal{O}(n^{-1}). \end{aligned}$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\begin{aligned} \sum_{j=0}^{n-1} \ln h_j = & -n^2 I_Q[\mu_Q] - \frac{n}{2} \ln(n) + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{2}{3} \ln \left(\frac{r_0}{r_1} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \\ & - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + \mathcal{O}(n^{-1}). \end{aligned}$$

THEOREM

Asymptotically on n ,

$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] - \frac{n}{2} \ln n + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) + F_V[A_{[r_0, r_1]}] + \mathcal{O}(n^{-1})$$

where

$$F_V[A_{[r_0, r_1]}] := -\frac{2}{3} \ln \left(\frac{r_1}{r_0} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr,$$

THEOREM

For the annulus case, where $r_0 > 0$, we have the following expression for the free energy asymptotically on n given by

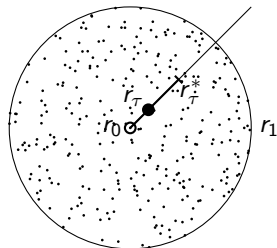
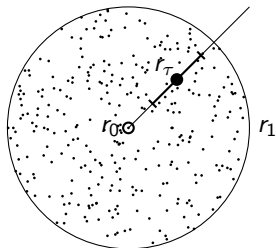
$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{n}{2} (\ln(2\pi) - E_Q[\mu_Q] - 2) + \frac{n}{2} \ln(n) + \frac{\ln(2\pi)}{2} + \frac{1}{2} \ln(n) + F_V[A_{[r_0, r_1]}] + \mathcal{O}(n^{-1})$$

where

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (sq'(s))' ds.$$

$$F_V[A_{[r_0, r_1]}] := -\frac{2}{3} \ln \left(\frac{r_1}{r_0} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr,$$



THEOREM

For the disk case, where $r_0 = 0$, we have the following, asymptotically on n , for the free energy

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2} (\ln(2\pi) - 2 - E_Q[\mu_Q]) - \frac{5}{12} \ln n \frac{\ln(2\pi)}{2} + \zeta'(-1) + F_V[D_{r_1}] + \mathcal{O}(n^{-1/12} (\ln n)^3).$$

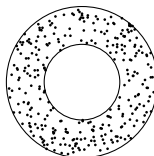
where

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (sq'(s))' ds,$$

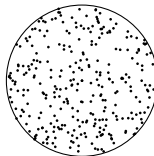
$$F_V[D_{r_1}] := \frac{1}{6} \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{3} \ln(r_1) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)} \right) - \frac{1}{32} \frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12} \int_0^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr.$$

For the annulus,



$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2} (\ln(2\pi) - 2 - E_Q[\mu_Q]) + \frac{1}{2} \ln(n) + \frac{\ln(2\pi)}{2} + F_V[A_{[r_0, r_1]}] + \mathcal{O}(n^{-1}).$$

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2} (\ln(2\pi) - 2 - E_Q[\mu_Q]) - \frac{5}{12} \ln(n) + \frac{\ln(2\pi)}{2} + \zeta'(-1) + F_V[D_{r_1}] + \mathcal{O}(n^{-1/12} (\ln n)^3).$$



for the disk

Obrigado!

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